Thread - 2025-02-22

https://x.com/RiikonenMarko/status/1893225968191291654

1/15 - 2025-02-22 09:06 UTC

Time to delve into the results of the "Juujärvi racing #131 display" on the night of the 19/20 November 2025 at the Ylikylä snowdump in Rovaniemi. These full fisheye images that I took in the beginning with Canon 80D serve as a preamble to the more depth stuff taken with cropped fisheye on Nikon D7000. I was just 50 meters from the snow guns and such close range means more variability in the ice fog. Bad for not allowing very long stacks, the instability turned out to be good for providing some insight to the halo relations of helic arc on the one hand and Moilanen arc on the other.

The images shown here are in the shooting order, they are either singe photos or stacks, the file numbers indicated. The lamp is above horizon in all images.

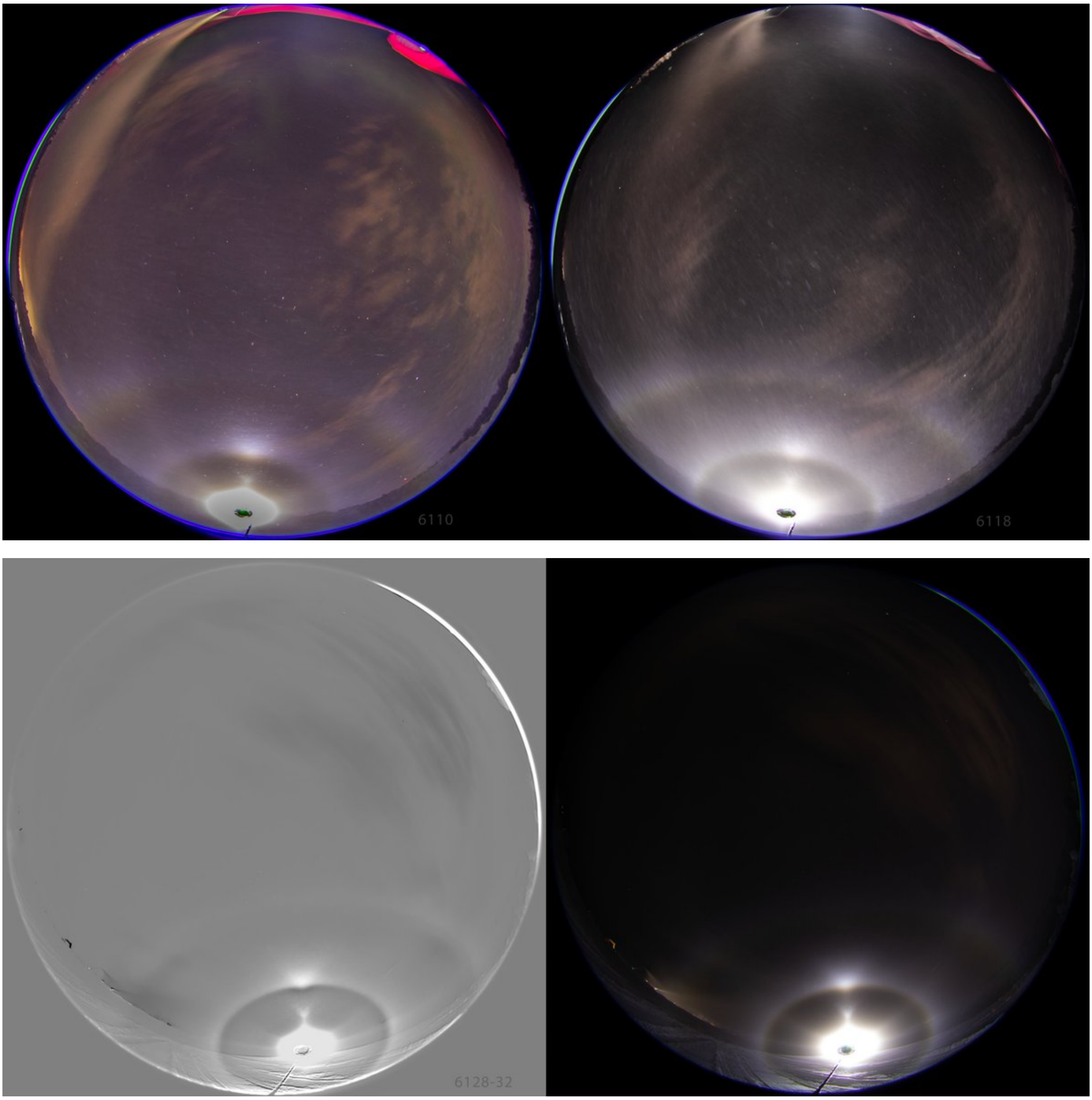
At first (6110, 6118) I had the lamp up on the dump-snow wall, the elevation maybe around five degrees. The uppervex Parry is lacking despite quite distinct helic arc, but then these are just single images. Opposite to the lamp is very broad diffuse anthelic arc / flip anthelic arc.

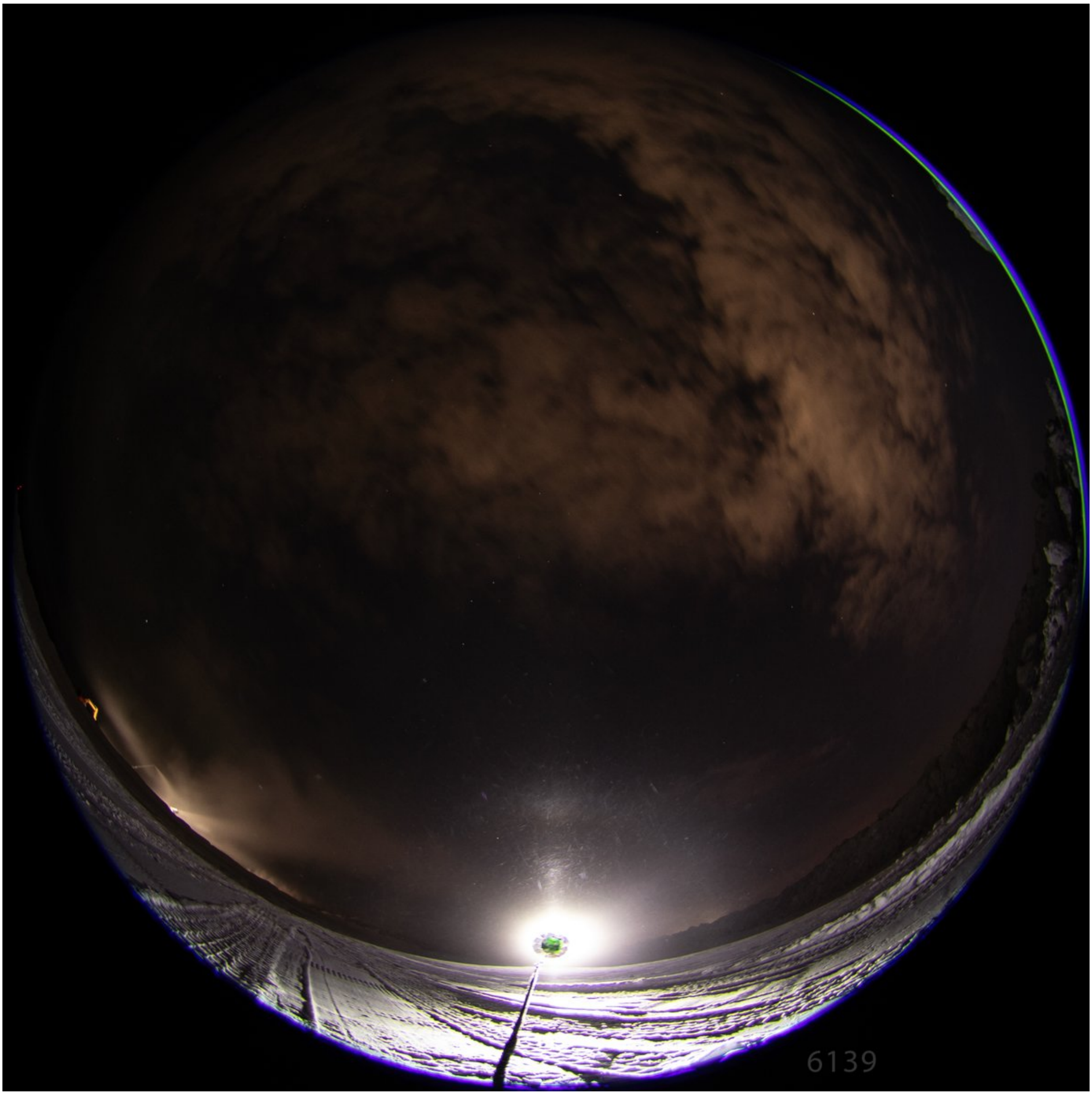
In the next images the lamp is a bit lower as I placed it on the roof of my car, in stack 6128-6132 maybe just 1 degs (this was my smaller Manfrotto, legs and central bar not extended), which may account for why Parry is not seen to accompany the helic arc as it is mixed up with the tanarc.

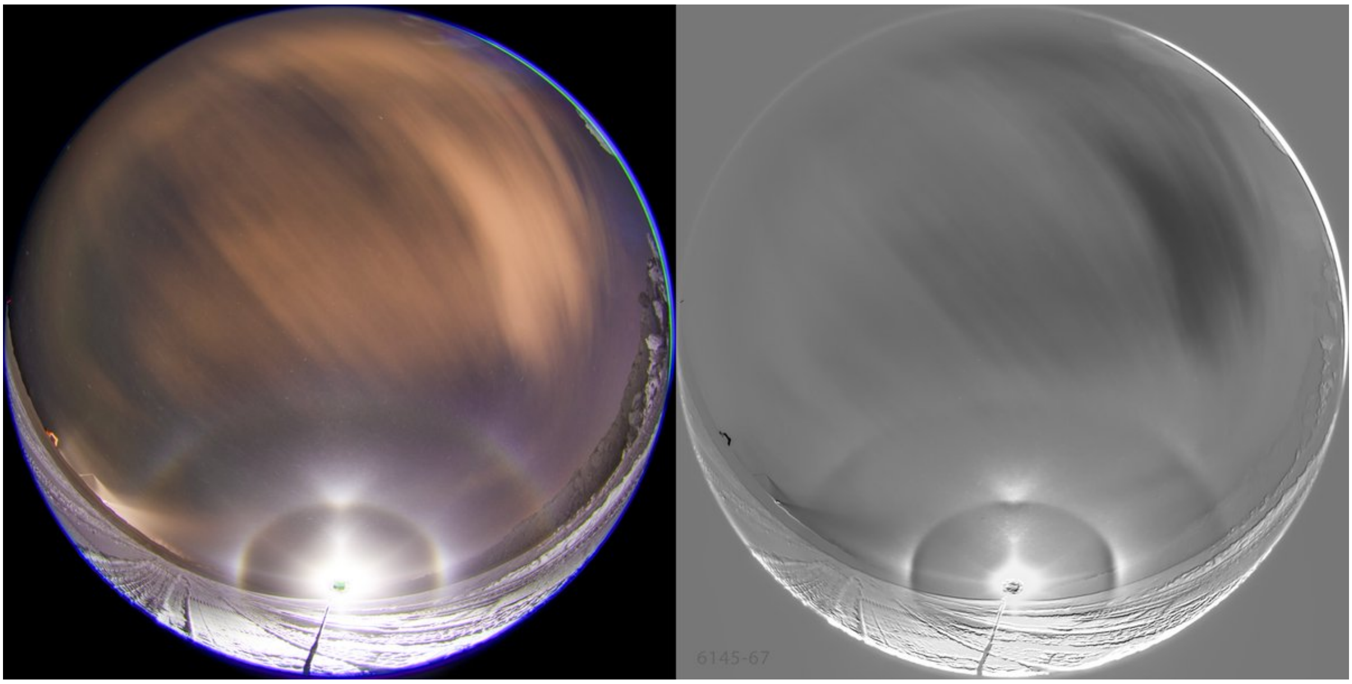
Then the varying display gave the single frame 6139, the first photo after I had changed the lamp on my bigger tripod, now fully extended on the car roof. This 10 seconds exposure has an unusual and interesting feature: Moilanen arc not accompanied by parhelia. Parhelia are a fixture of the Moilanen displays, perhaps suggesting an intimate connection: that Moilanen crystals make also parhelia. Alternatively Moila crystals don't make parhelia but the conditions that spawn those crystal also involve as a rule the formation separate population of simple plates. Answer to this question is of some importance as we try to crack the mystery of the Moilanen arc. I have known only one Moila display without parhelia, the display by Bertram Radelow in Davos, Swizerland, on 2 December 2010: https://t.co/bJ1K19m6mM But two single shots in clumpy diamond dust hardly constitutes a compelling evidence – could be that at the moment of those photos different kind of crystals from those at the Moilanen location were present at the parhelia location. The left side photo actually may have parhelion, the 22° halo is a bit enhanced on the side.

And even if in the much smaller volume of spotlight beam the whole display is great deal more likely to be made of uniformly composed ice fog, I wouldn't put too much weight on one 10s shot happening to have Moila without parhelia (the next shot has already parhelia developing). But given later with Nikon I got several successive shots of parhelialess Moilanen, that starts already sharpening my ear for the prospect of Moila crystal indeed not making parhelia.

The last image (6145-6167) is a more lengthy stack and has Parry arc to accompany helic arc. The distance between Parry and tanarc indicate lamp elevation of 4-5 degrees.

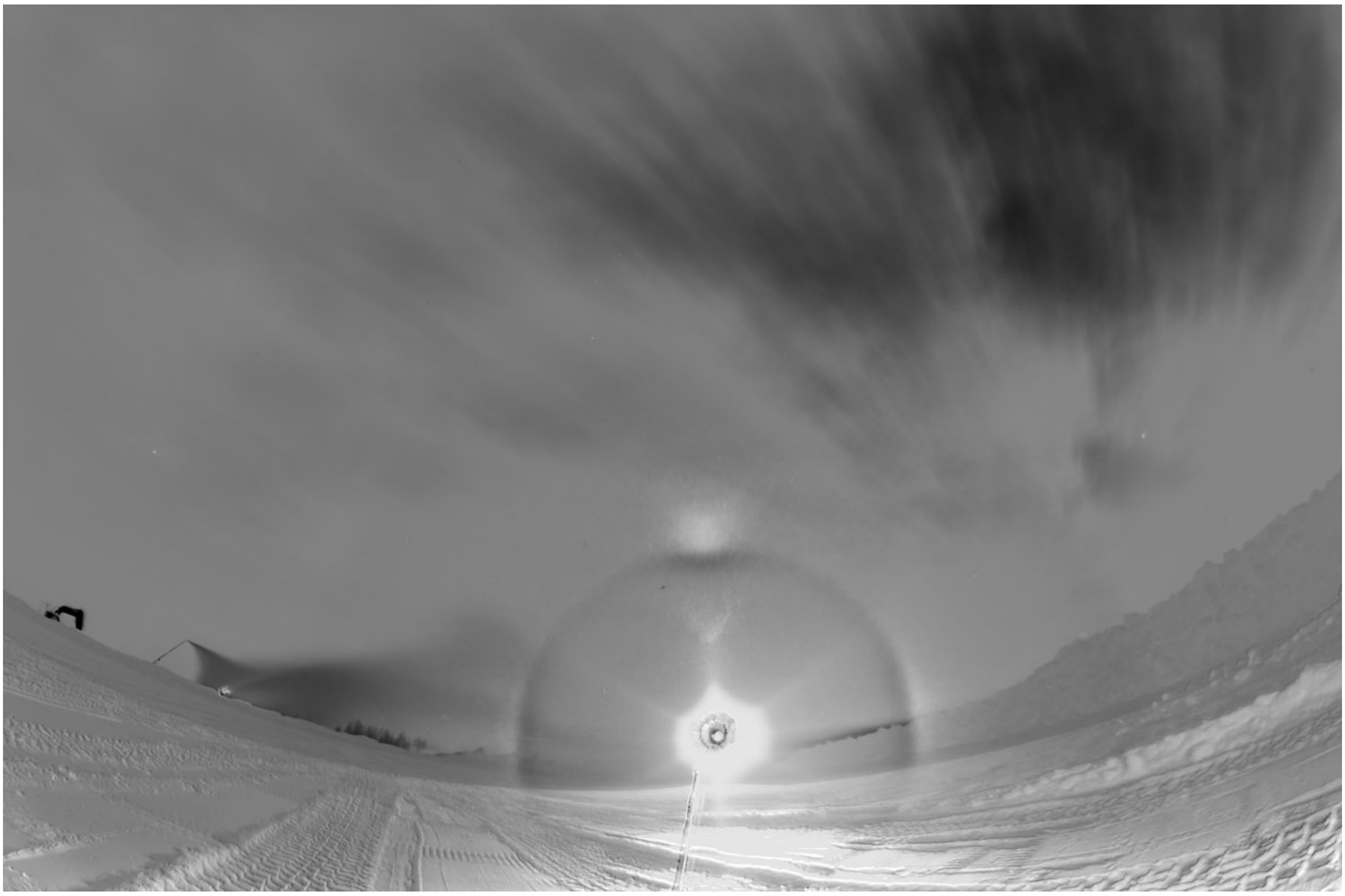

6110
6118
6128-32

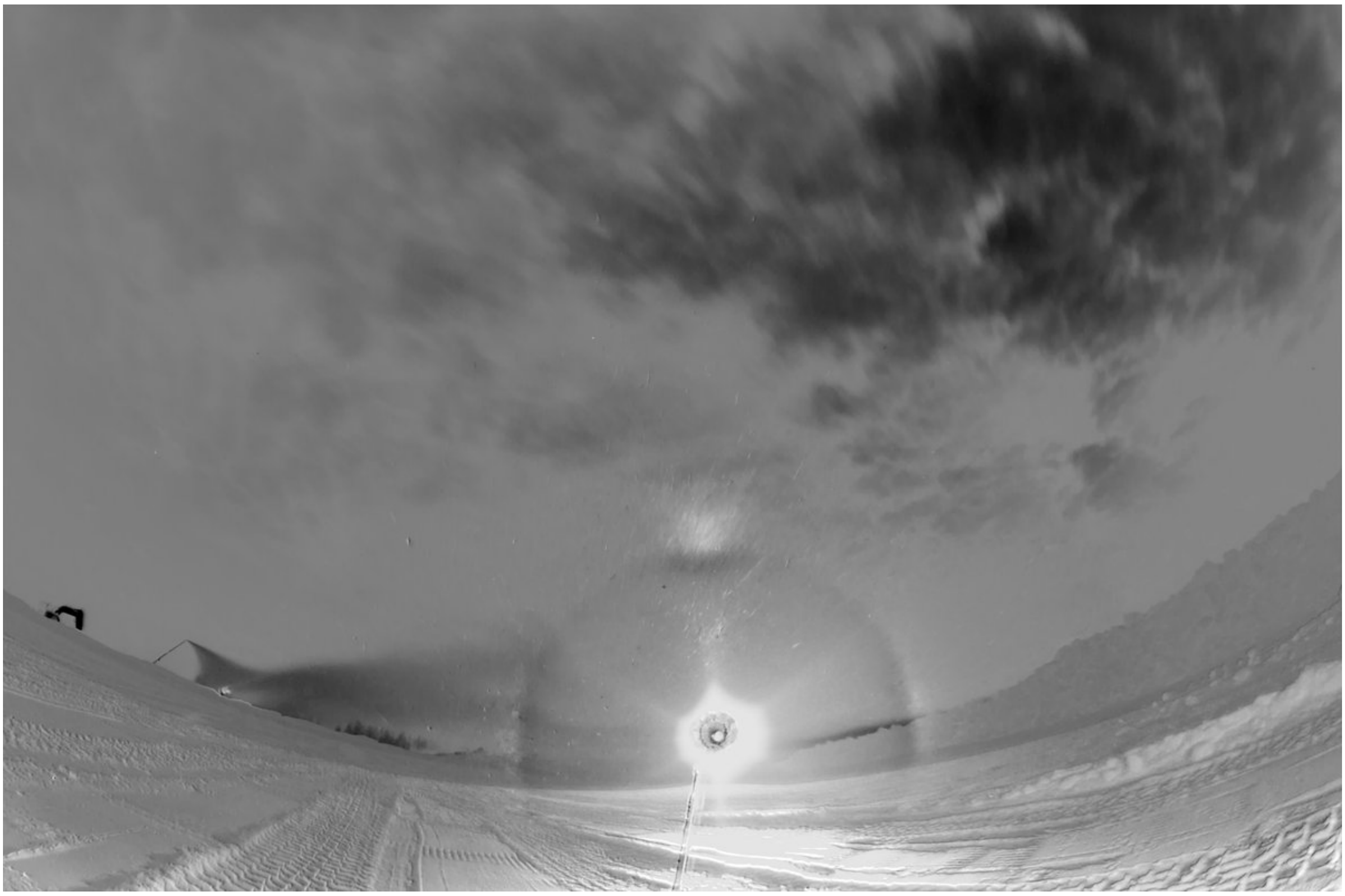

6139



2/15 - 2025-02-26 06:08 UTC

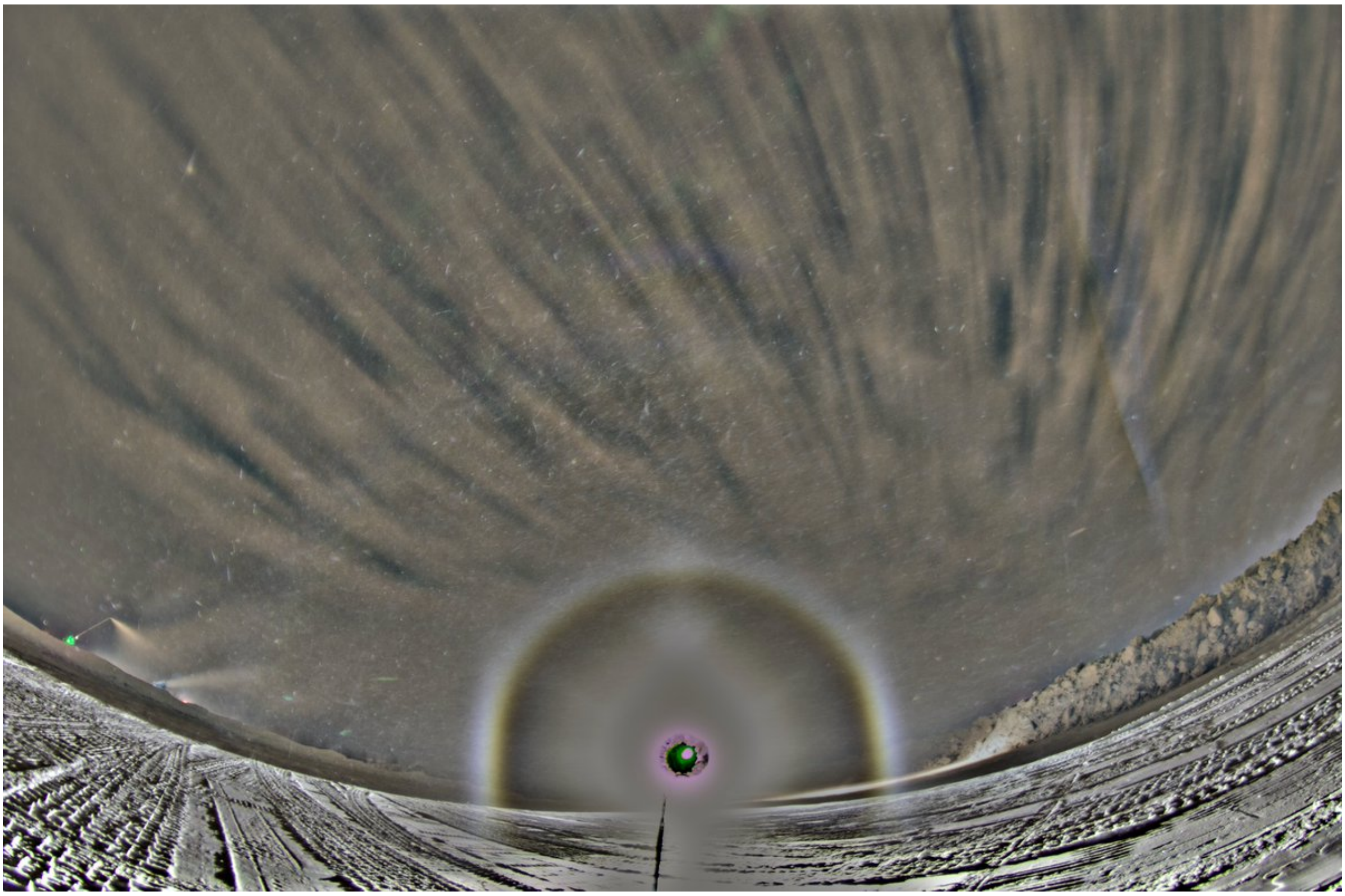
15/16 February 2025 Rovaniemi. The more depth results of the Juujärvi racing #131 display from my other camera, Nikon D7000. I have made stacks of the steady stages of the display, starting with this 5 frame stack 4015-4019 which is the only time with this camera showing clearly uppervex Parry to accompany the pretty much ever-present helic arc. Uppervex is actually visible in the last single frame 4019, as shown by the two last images. Exposures 15s.

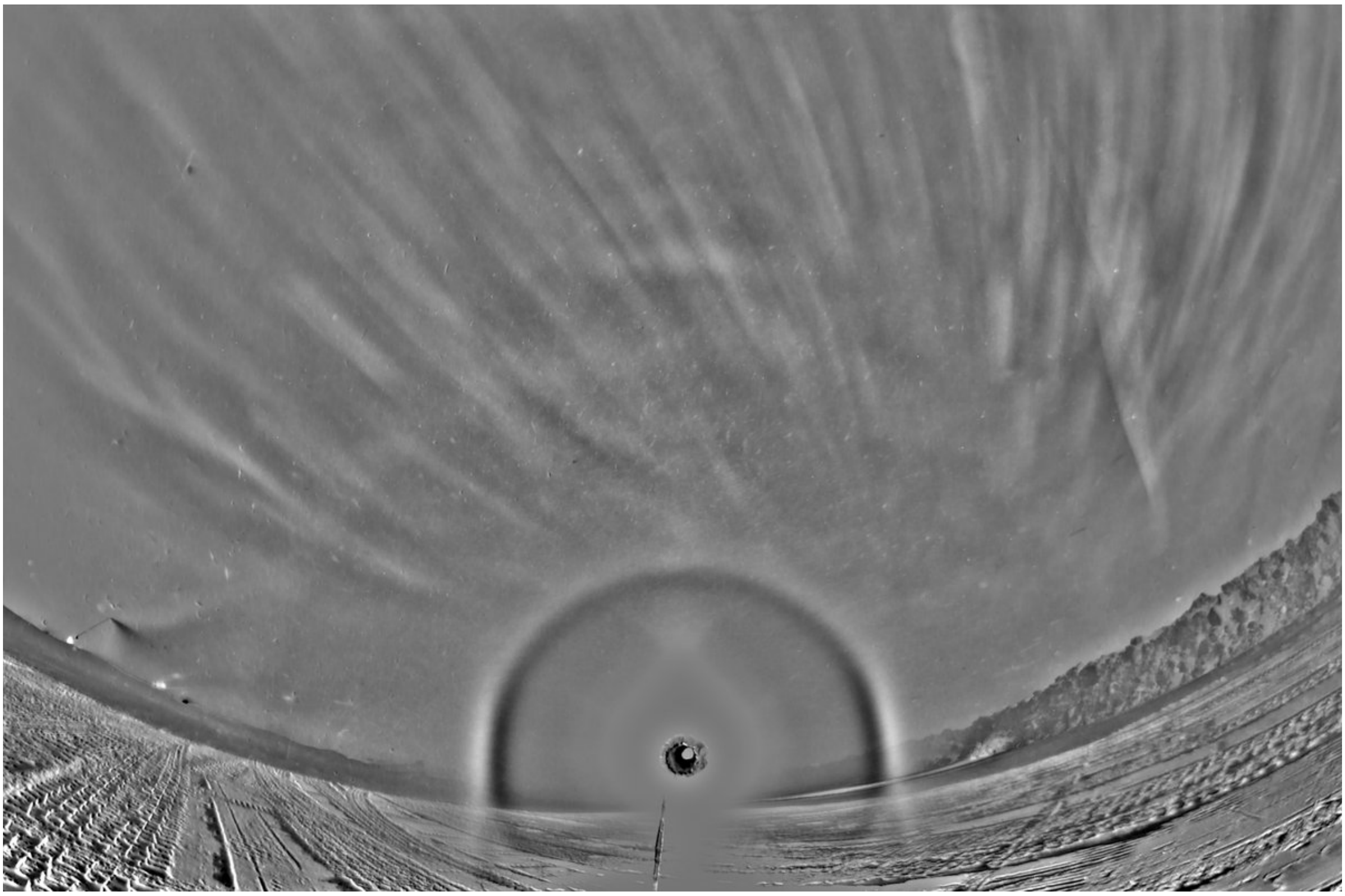




3/15 - 2025-02-26 06:12 UTC

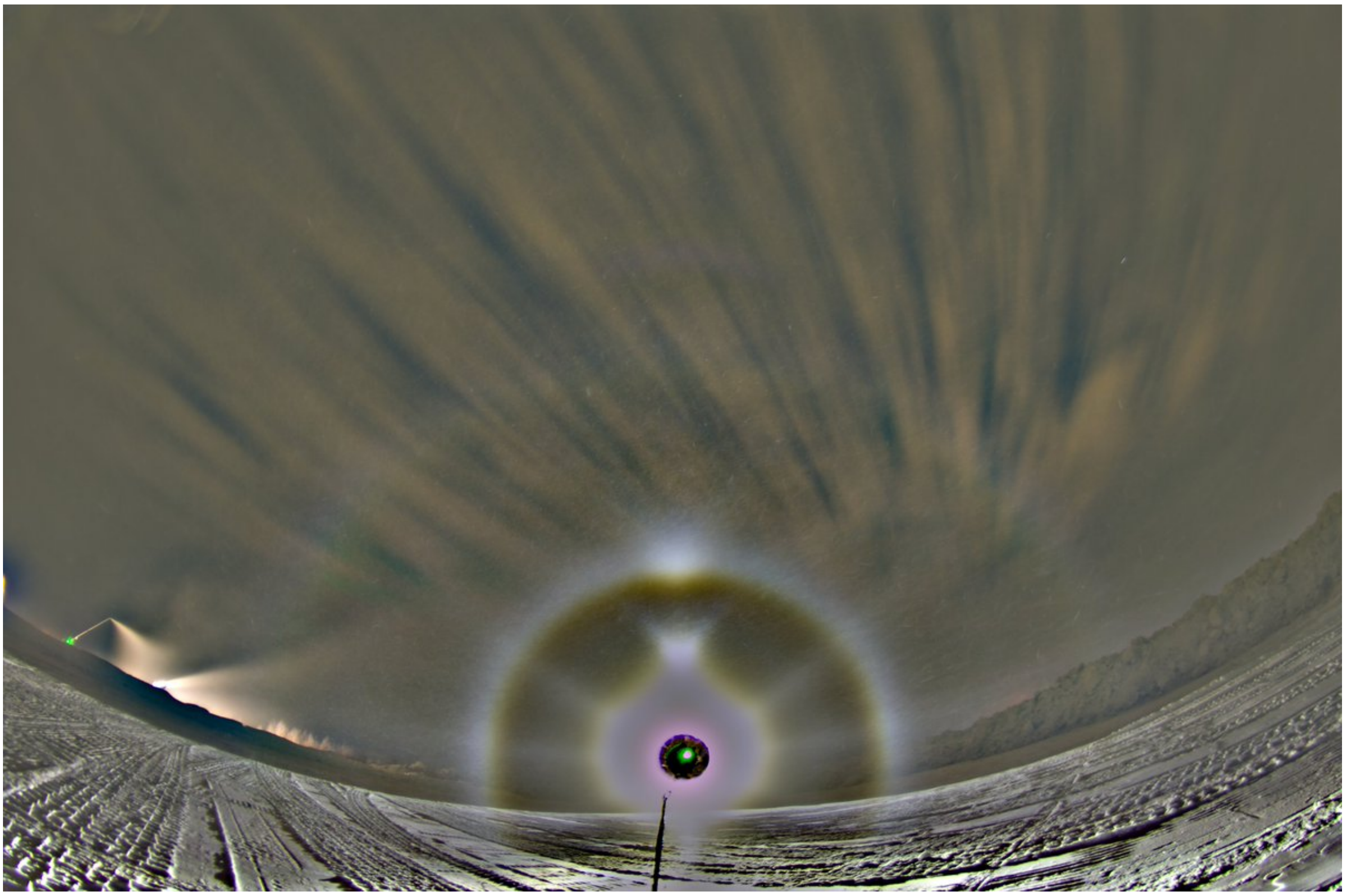
15/16 February 2025 Rovaniemi. 4055-4060 6f. 15s exposures.

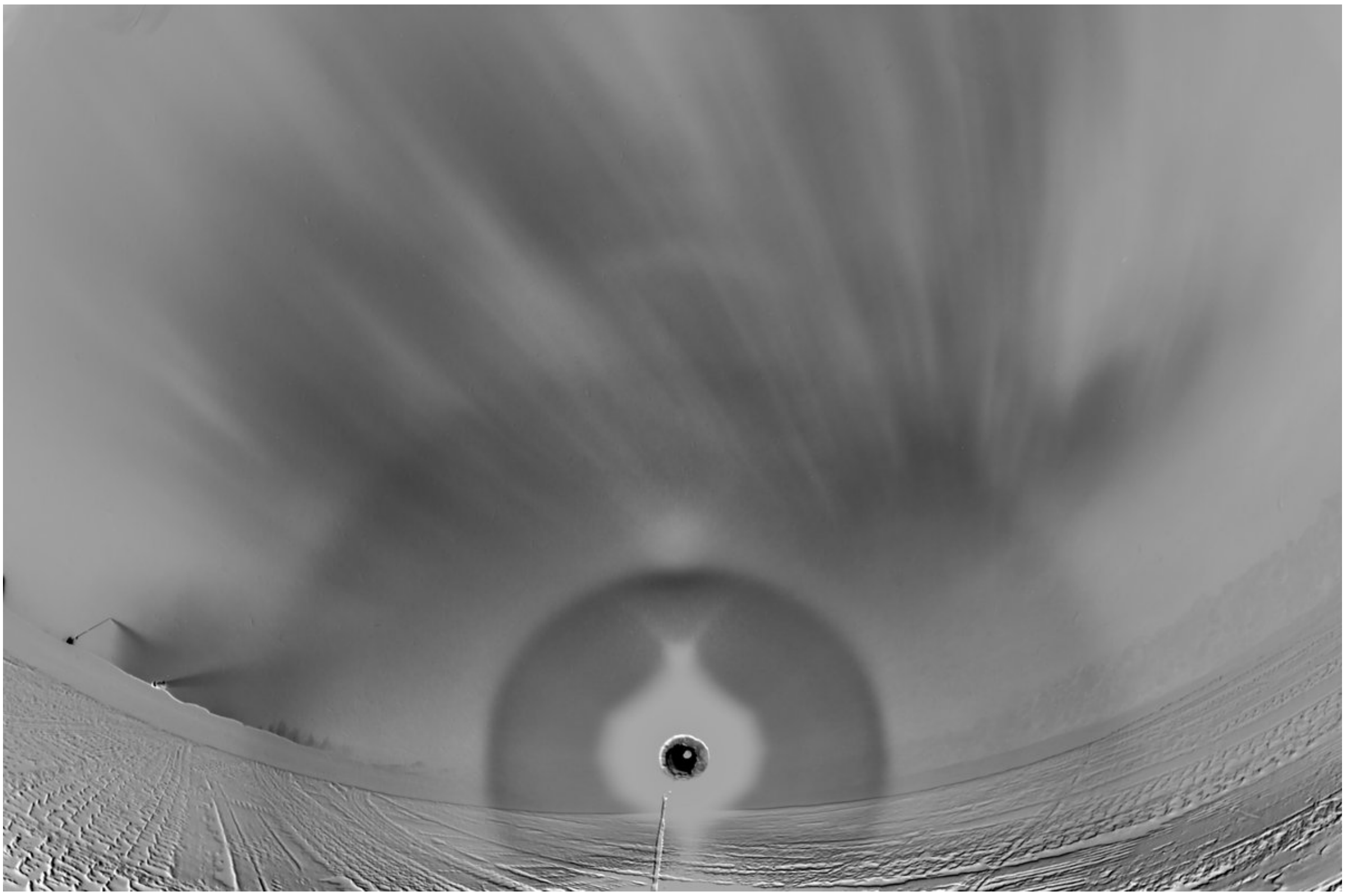




4/15 - 2025-02-26 06:15 UTC

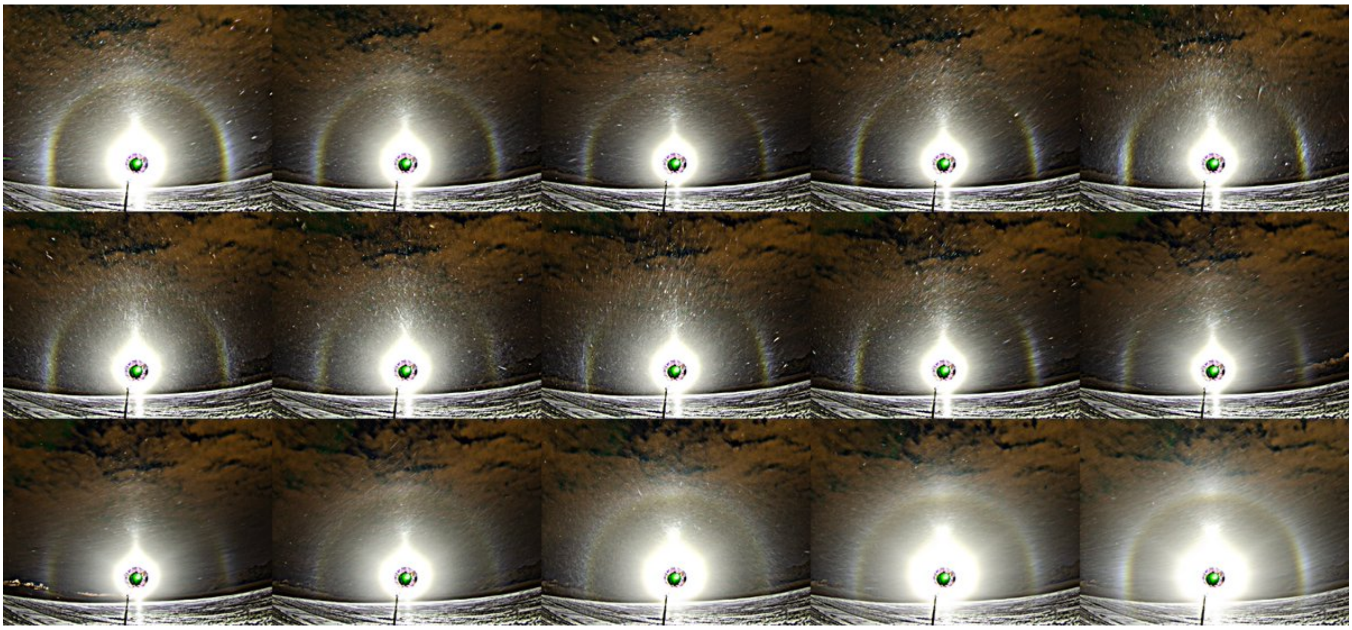
15/16 February 2025 Rovaniemi. 4076-4091 16f. The greyscale images are br and bg. Maybe they have a suggestion of Parry. Exposures 10s.





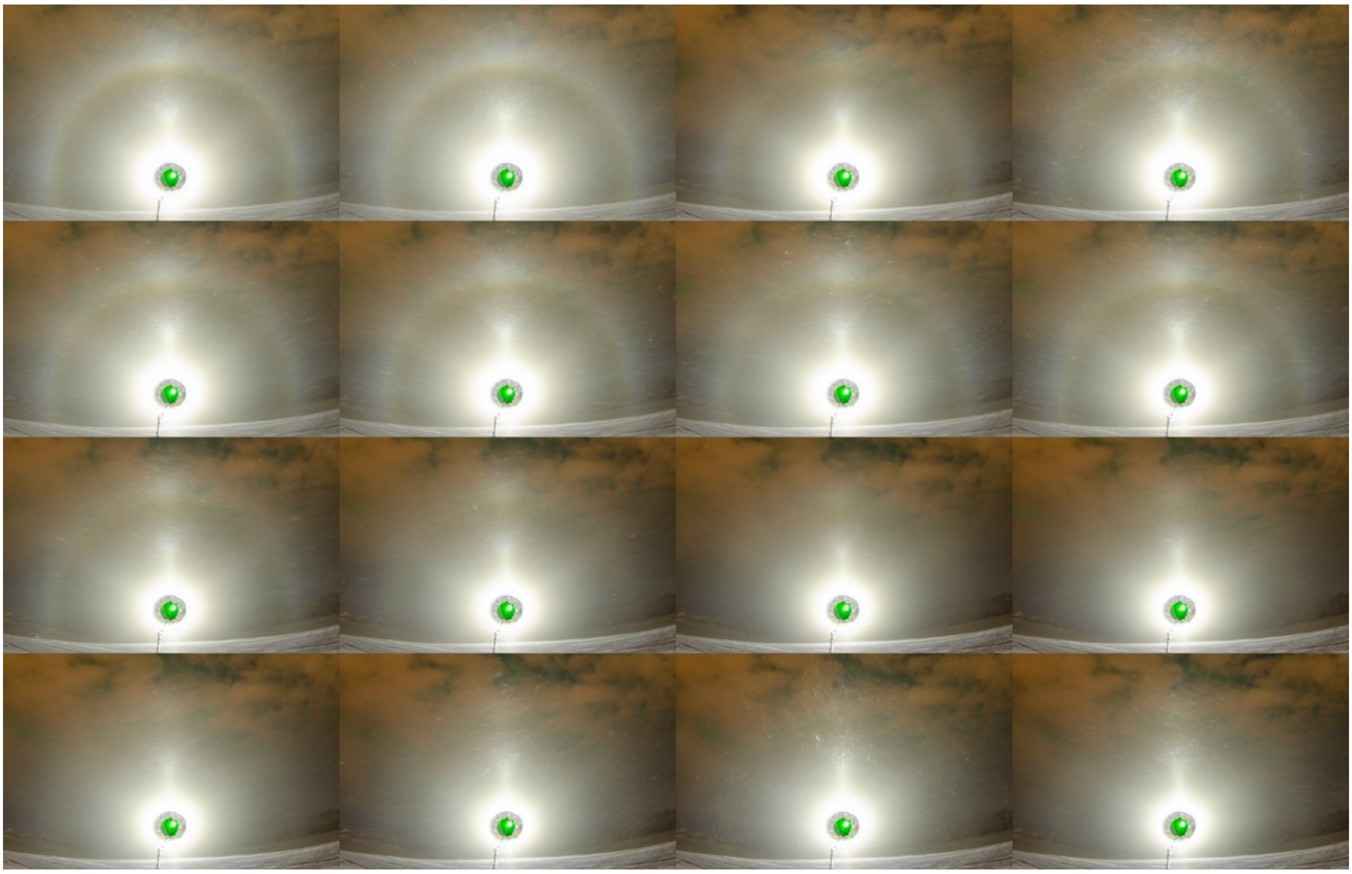
5/15 - 2025-02-26 06:17 UTC

15/16 February 2025 Rovaniemi. Here are one time usmed single frames 4062-4076 in between the two previous stacks. Exposures 10s.



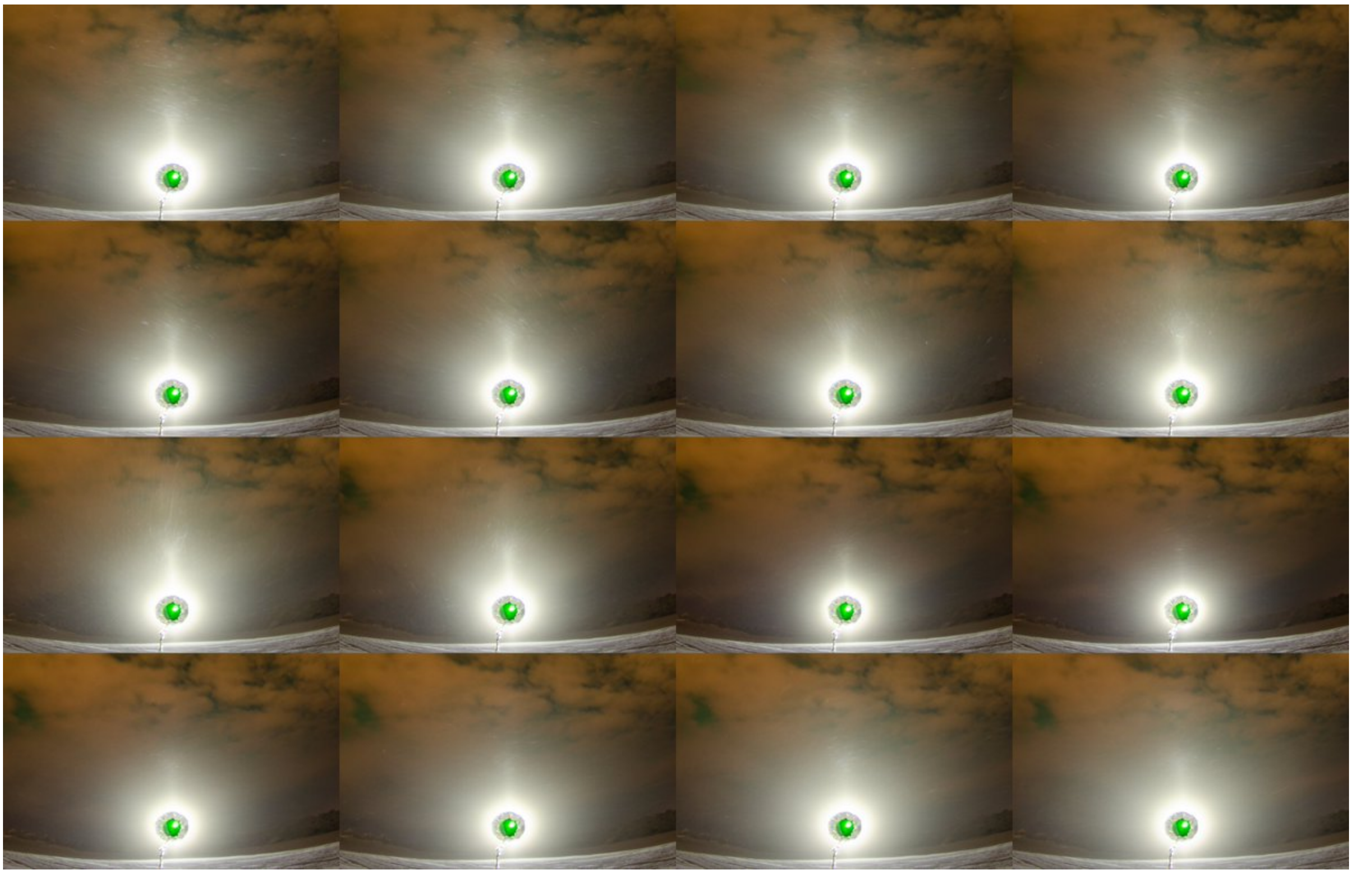
6/15 - 2025-02-26 06:26 UTC

15/16 February 2025 Rovaniemi. Then came the night's paydirt: Moila started showing up for significant stretches without much at all parhelia to talk about. This 4092-4107 frame collash is the start of it. I went light on the image enhancements, just the auto button in Photoshop Camera Raw. Exposures 10s.



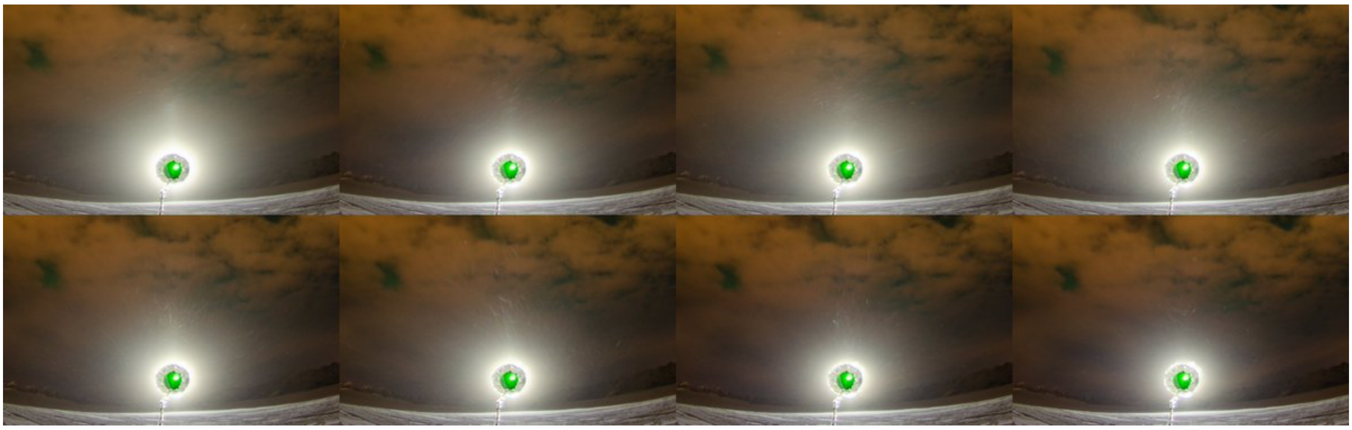
7/15 - 2025-02-26 06:27 UTC

15/16 February 2025 Rovaniemi. 4108-4123. Exposures 10s.



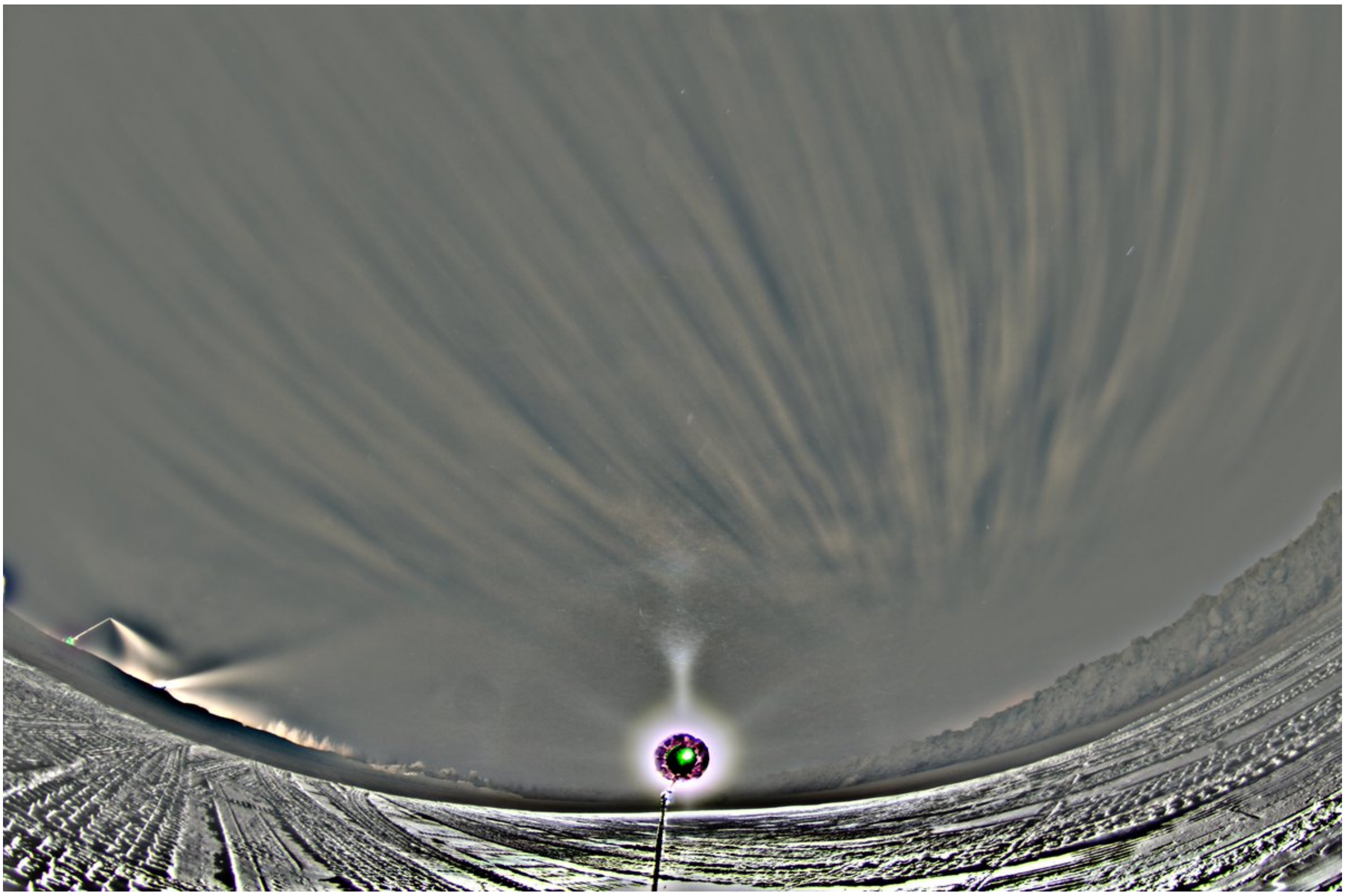
8/15 - 2025-02-26 06:28 UTC

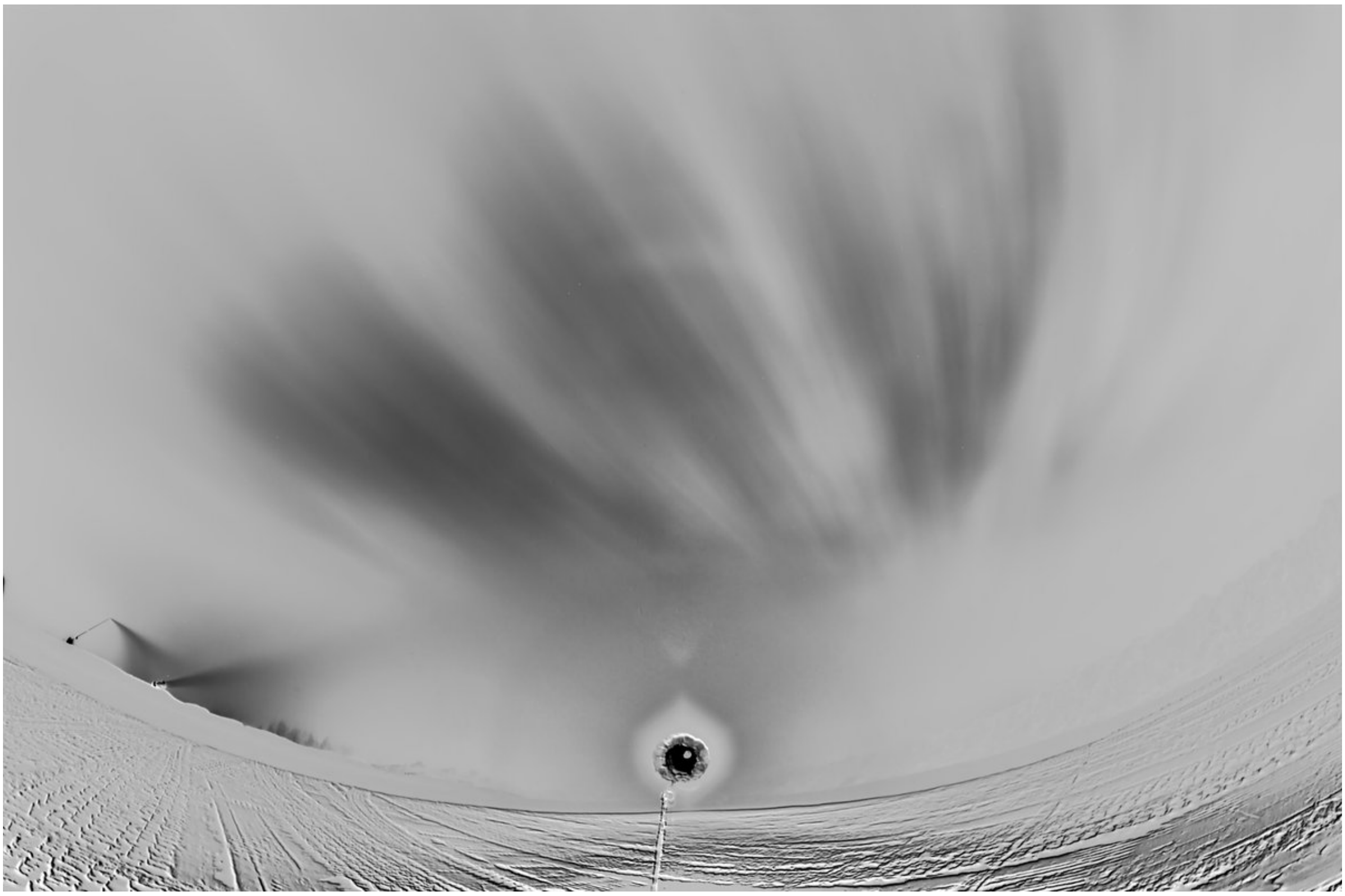
15/16 February 2025 Rovaniemi. 4124-4131. Ice fog vanished for a while and I stopped taking photos. Exposures 10s.



9/15 - 2025-02-26 06:38 UTC

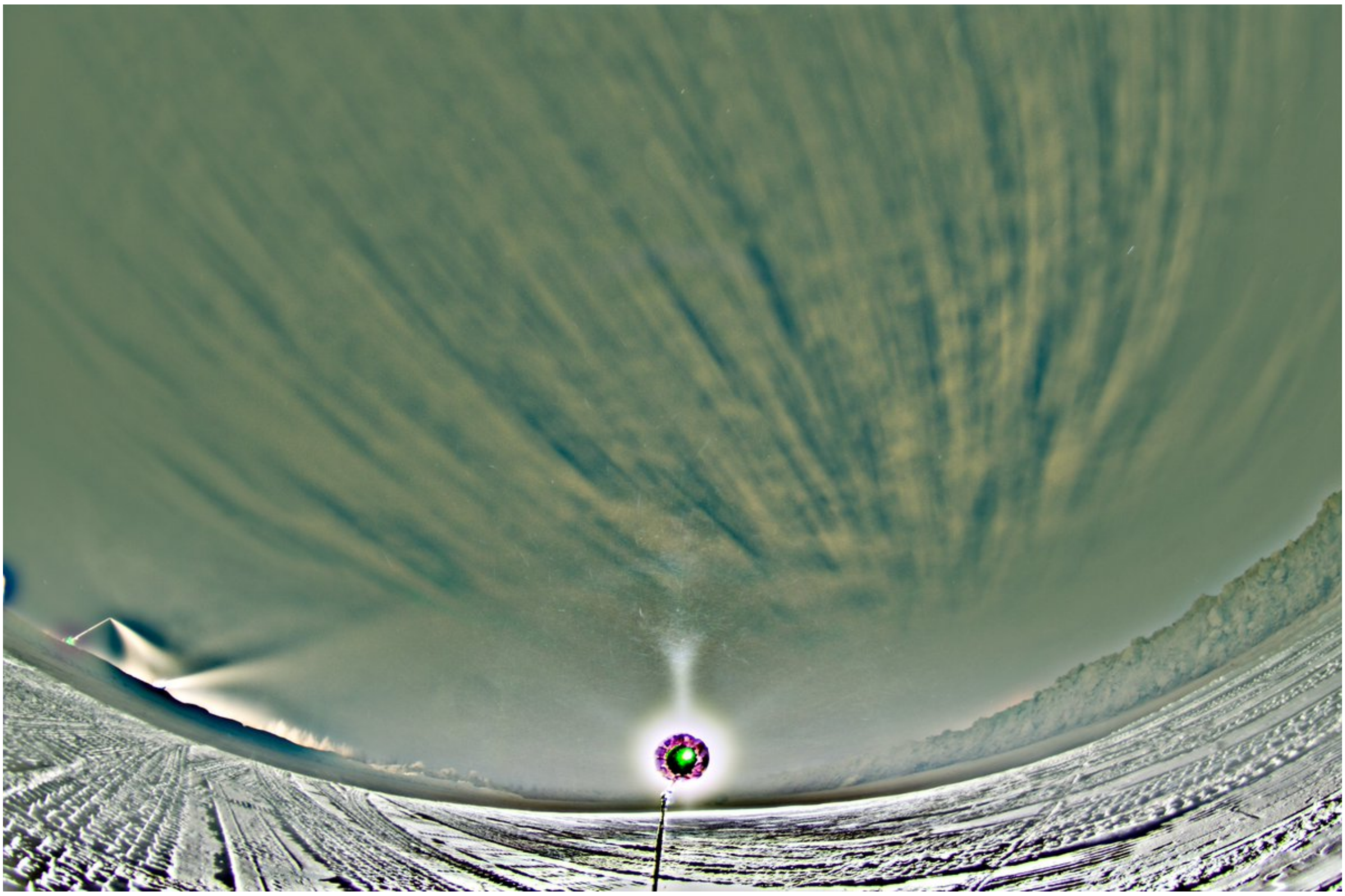
15/16 February 2025 Rovaniemi. 4101-4124 24f. This stack is the stretch where the parhelia presence was at minimum.

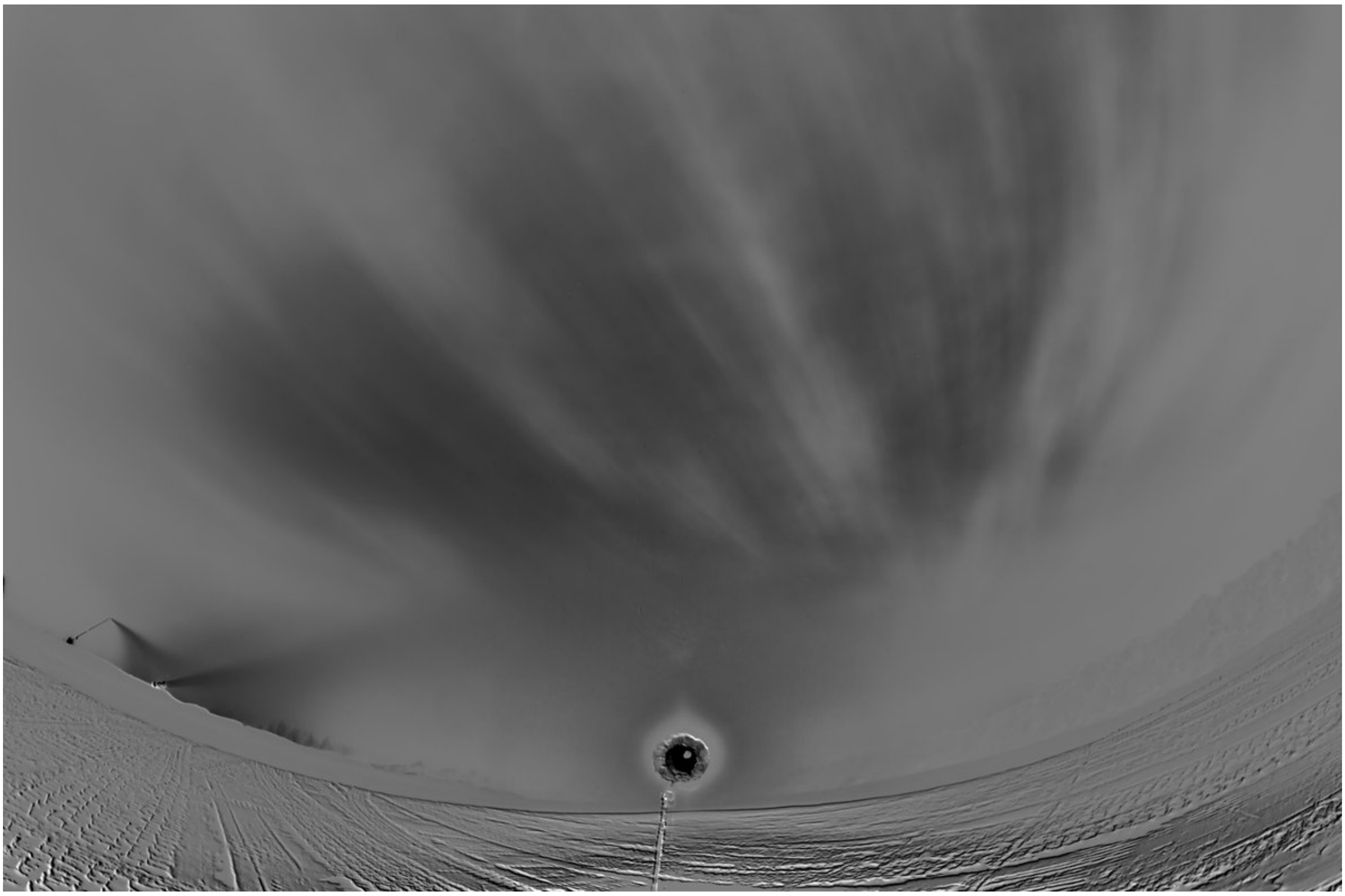




10/15 - 2025-02-26 06:40 UTC

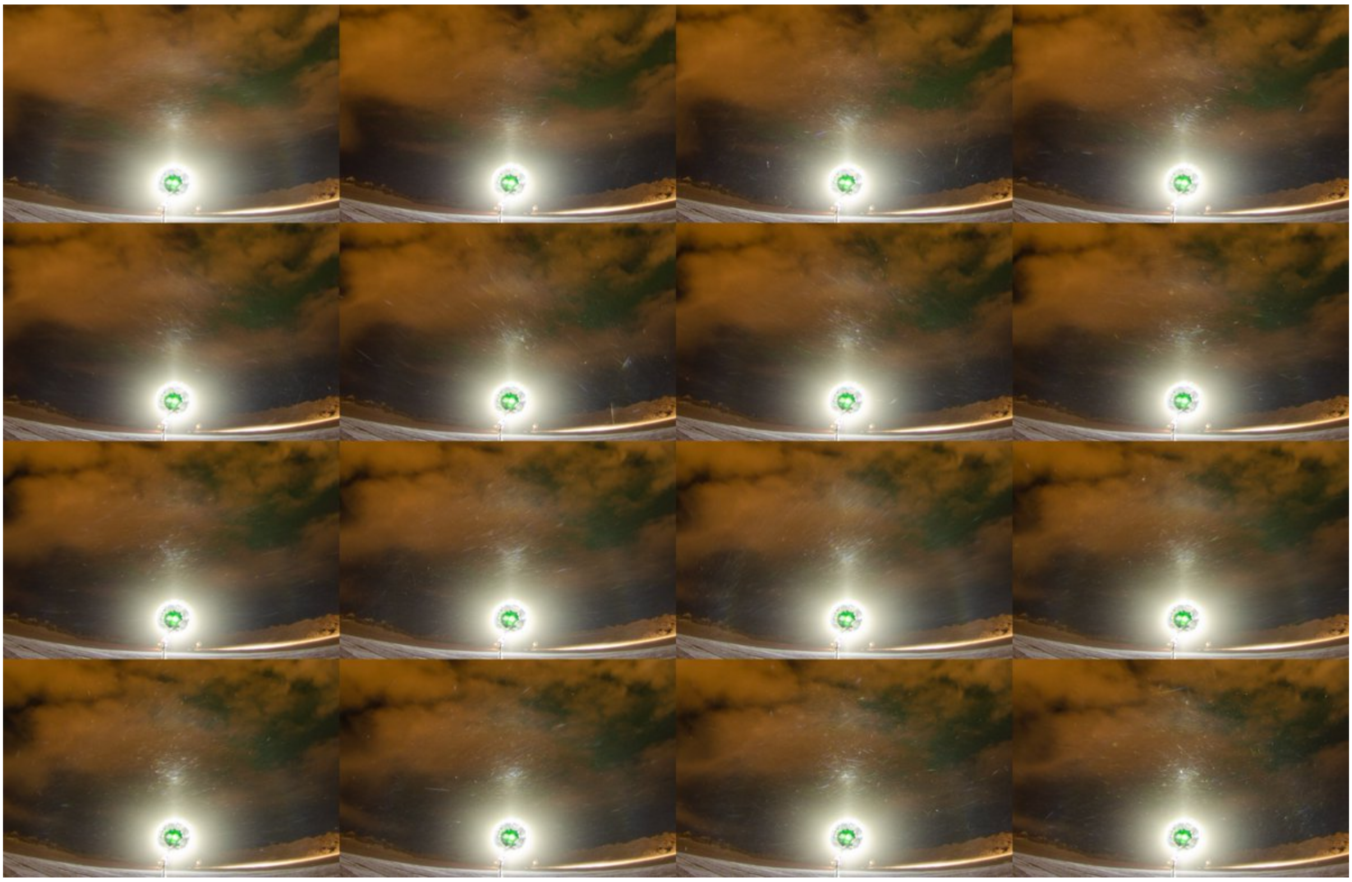
15/16 February 2025 Rovaniemi. Here is have cherry picked from the previous stack those frames that didn't seem to have parhelia at all, the frames 4104, 06, 08, 11, 12, 15-24. Something still comes up on the sides.





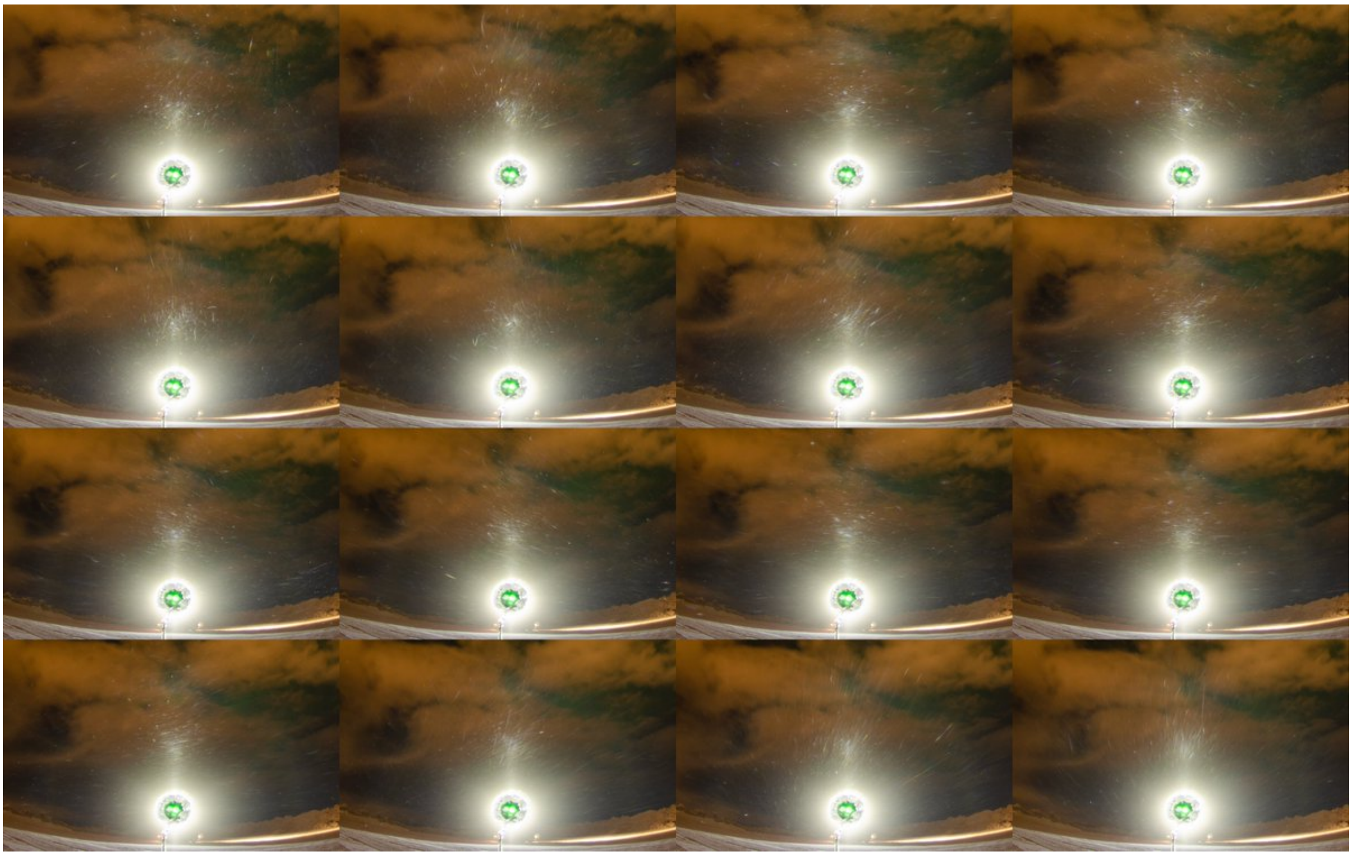
11/15 - 2025-02-26 06:41 UTC

15/16 February 2025 Rovaniemi. 4132-4147. When the ice fog came back, there was another spell with Moilanen occurring with little, or not at all, parhelia in these 10s frames.



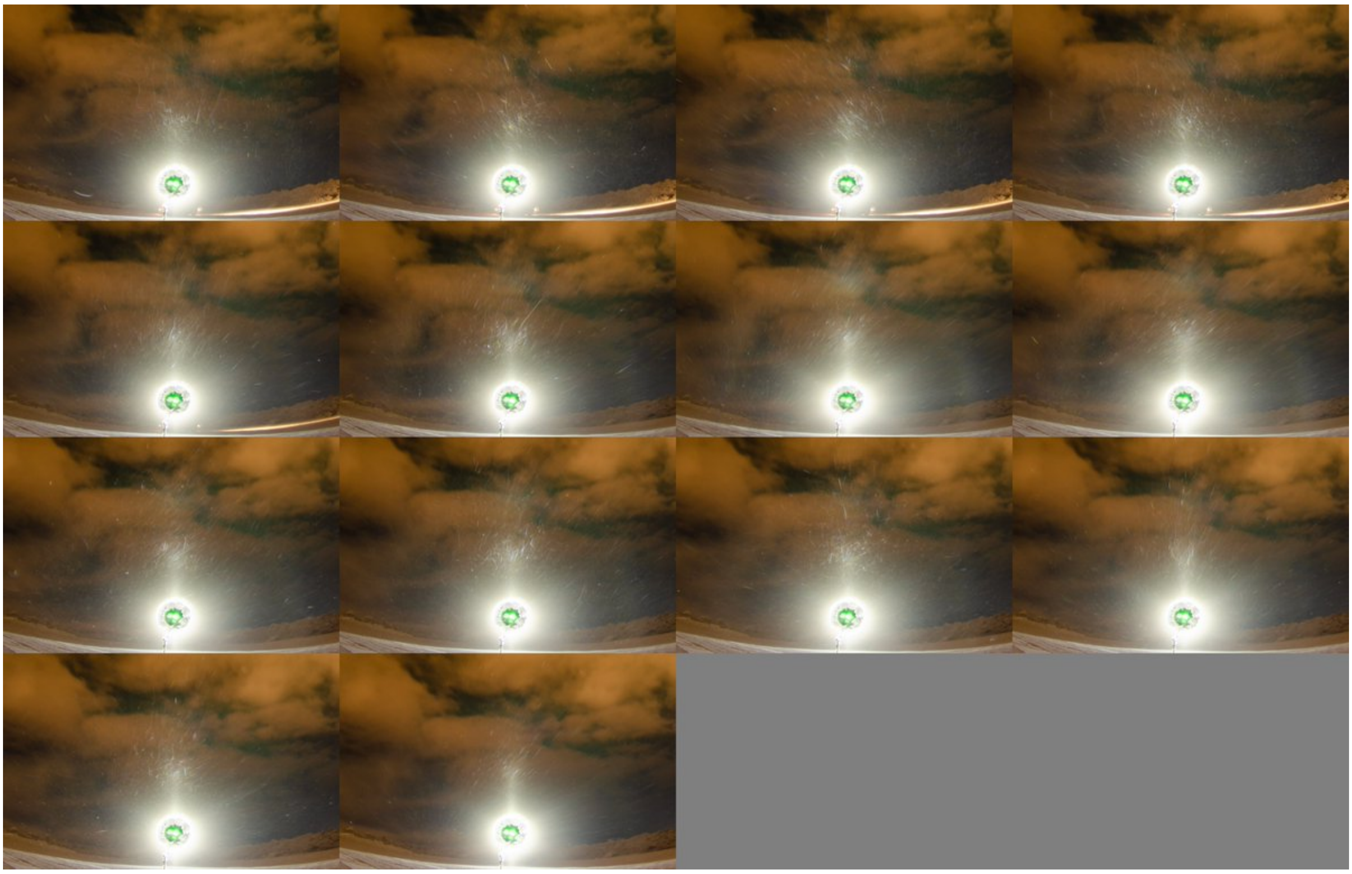
12/15 - 2025-02-26 06:42 UTC

15/16 February 2025 Rovaniemi. 4148-4163.



13/15 - 2025-02-26 06:45 UTC

15/16 February 2025 Rovaniemi. 4164-4177.



14/15 - 2025-02-28 06:34 UTC

15/16 February 2025 Rovaniemi. Here is s stack of all 45 frames of the three posts above, 4133-4177.

So where this all leaves us? When stacked, parhelia came up always in this display, however faint. Even if there was a display with absolutely no parhelia presence, it could not be taken as proof that Moilanen crystals don't make parhelia. As we all know, crystal quality affects how much halos from given orientation can be seen. For example, in the spotlight display that Mikkilä, Moilanen and I observed in Rovaniemi on the evening of 22 November 2015, 46° latarc is present only as the faintest whiff despite the well fleshed tanarc: https://t.co/LLFBiQSpZq

By the same token a paranoid mind can't rule out Moilanen crystal making Parry arc, even if there are much more Moilanen arcs without Parry. Here is a sample of such cases from Taivaanvahti archives:

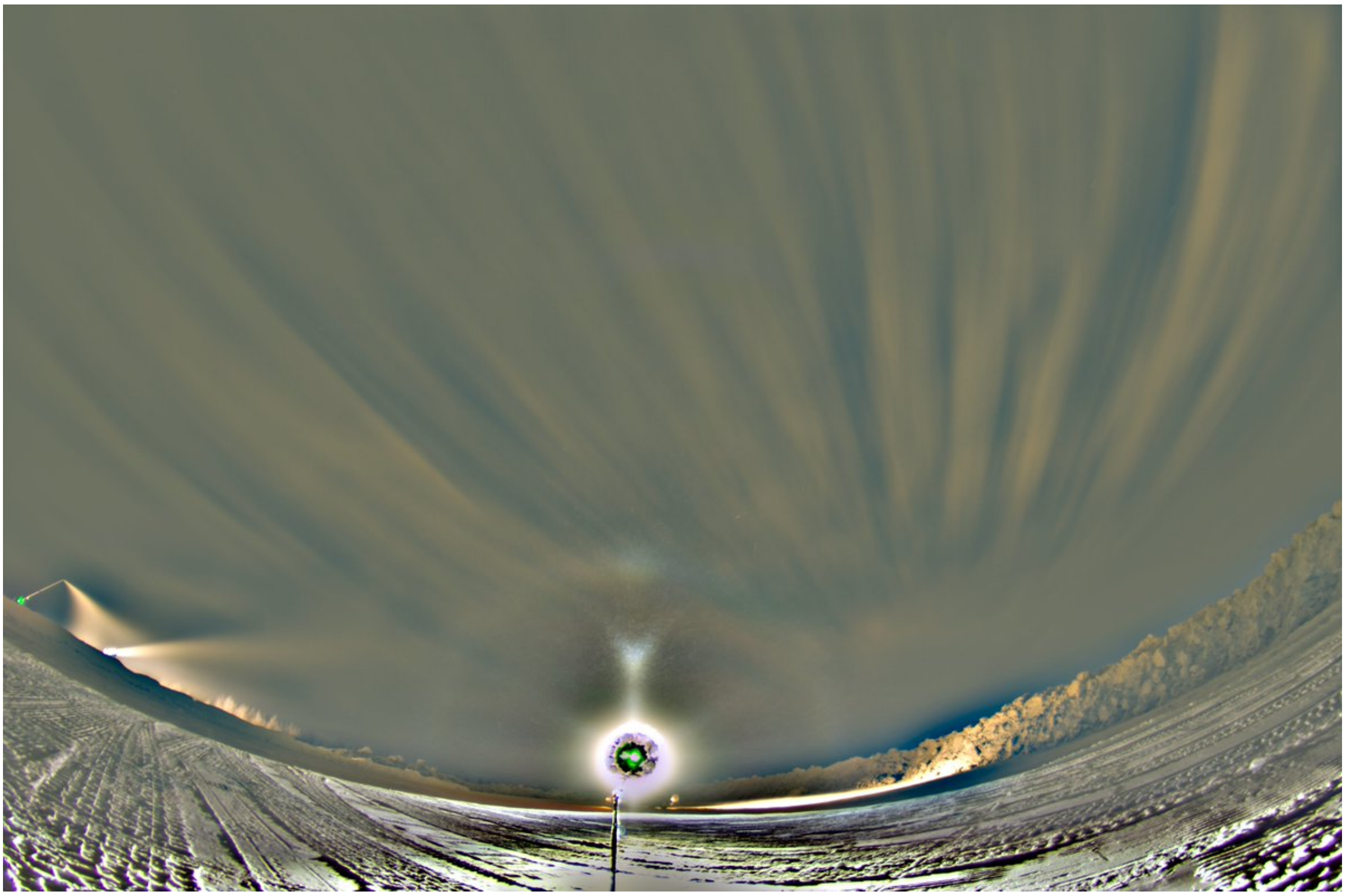
https://t.co/l0jQBku54y
https://t.co/Dpcy6qfzlS
https://t.co/L86Jn1Qc8k
https://t.co/7WAYbqyGS7
https://t.co/WnXzMGzNFi
https://t.co/YSZ6wXnqcU
https://t.co/qL5K89P79W
https://t.co/M42cr0W3s4

I selected only stacked displays and moreover only those displays which do not have much of the tanarc brightening either. Many of these contain also the "alternate Moilanen", stacking probably has helped to make it materialize.

Of course discussion of Moilanen arc related halos needs must involve crystal samples. On the basis of crystals sampled ice fog displays, it appears that Moilanen arc swarms contain exclusive crystals: crossed elongated plates. The sample that Jukka Ruoskanen took of one display late December 2015 provides a fine example: https://t.co/klk3NNfv5t If the prism responsible for M arc runs along the elongated direction of these crystals, that explains nicely why we see long wings in Moilanen arc instead of a stub like 9° plate arc.

The alternate Moilanen arc throws its own spanners in the works on any Moilanen arc formation considerations. It seems to form from the same 34 degree angle as Moilanen arc, but instead the prism pointing straight up which is how Moilanen arc forms, the prism for alternate Moilanen has the other face vertical. So which is right? That alternate Moila is made by normal Moilanen crystals in an alternate orientation, or that the orientation is the same but the prism itself has grown tilted?

The helic arc - Parry arc relationship in this display seems a matter of intensity: Parry is visible only when the helic arc is at its brightest. Usually even if Parry is not visible in helic arc containing display, there is the top brightening on 22° halo. Here is an example of helic arc in a display which does not seem to have even that: https://t.co/ACnWF9fagL True, the flipsun is disturbing, but its effect is reduced in br image. Thus in this display no other halo than helic arc indicates C-axis horizontally oriented crystals. There are more extreme examples: Jukka Ruoskanen has a photo of helic arc appearing only with pillar.



15/15 - 2025-03-01 10:45 UTC

The date is wrong in most posts above, should be 19/20.